

Data Challenge Report

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Overview

I applied HypotheSAEs (Movva et al., 2025) to a new domain: U.S. job postings linked to firm-level financial outcomes. My dataset contains job advertisements posted between March 1–9 2022 by S&P 1,500 firms, together with each firm’s return on assets (ROA) in 2020 and 2021. I defined a binary label equal to 1 if a firm’s ROA declined from 2020 to 2021 and 0 otherwise. The task aims to ask: what latent features of job-posting text are associated with whether the posting comes from a firm whose financial performance declined?

I used HypotheSAEs to embed each posting, train a sparse autoencoder (SAE), identify SAE features whose activations were correlated with the binary ROA-decline label, and then use an LLM to describe those features in natural language.

The pipeline worked as intended: the SAE learned coherent and interpretable features, and several features were strongly associated with the label in the training data. However, the strongest features mainly captured differences in the types of jobs and industries represented in the postings (e.g., financial-services, trucking, and retail roles) rather than general differences in how firms described otherwise similar jobs. This revealed a limitation of using HypotheSAEs with hierarchical data: when the label is defined at the firm level but the text is observed at the job-posting level, predictive SAE features may capture between-firm differences in occupational/industry composition rather than the recruiting-language dimensions of interest (e.g., masculine versus feminine wording).

Data and Preprocessing

The dataset contains 16,495 U.S. job postings from 307 S&P 1,500 firms. The job postings were collected between March 1 and March 9, 2022, and include the firm name, job title, and full posting text. I linked these postings to firm-level financial data measured at the end of fiscal years 2020 and 2021. I then defined a binary firm-level outcome equal to 1 if a firm’s year-end return on assets (ROA) declined from 2020 to 2021 and 0 otherwise. 47.6% of firms experienced an ROA decline.

A key preprocessing issue was that the outcome is measured at the firm level, while the text is measured at the job-posting level. To avoid leakage, I split the data by firm rather than by posting. Firms were stratified by the ROA-decline indicator and assigned to training, validation, and holdout sets in approximately a 70/15/15 split. This produced 214 firms (14,274 postings) in training, 46 firms (950 postings) in validation, and 47 firms (1,271 postings) in the holdout set, with no firm appearing in more than one split.

I also cleaned the posting text before embedding. I removed URLs, email addresses, and company names. Because company names often appeared in abbreviated or branded forms, with the help of Codex, I additionally created aliases for firms with at least 10 postings, covering 88 firms in total (e.g., both “Cracker Barrel Old Country Store” and “Cracker Barrel” were removed from the relevant postings); for the remaining firms with fewer than 10 postings, I removed the firm name as recorded in the dataset. I retained job-related content, including occupation names and descriptions in the posting, which later became important for interpreting the discovered features.

SAE Training and Feature Interpretation

I first embedded each cleaned job posting using OpenAI’s `text-embedding-3-small`, which produces a 1,536-dimensional vector for each text. Because the posting texts were long, the default embedding pipeline exceeded the API’s per-request token limit. I therefore reduced the batch size, processed the corpus in smaller chunks, and cached the embeddings locally to avoid repeated calls.

I then trained a sparse autoencoder using the same general setup as the repository’s Quickstart notebook. I used $M = 256$ latent features, sparsity level $K = 8$, and Matryoshka prefix lengths of 32 and 256. Training stopped early after 48 epochs, with a final dead-neuron ratio of 0.004.

To check whether the learned SAE features were interpretable, I randomly sampled 5 neurons and asked the interpreter model to describe the textual pattern associated with each one. The resulting descriptions were coherent and easy to recognize. Examples included Colorado-specific job information, Equal Opportunity Employer language, and freight or logistics-related work.

Following the correlation-based option in the Quickstart notebook rather than the L1-regularized selection used in the paper, I ranked SAE features by the correlation between their activation values and the binary ROA-decline label. I selected the 20 most predictive features and used the language-model interpreter in HypotheSAEs to generate a natural-language description for each feature.

Predictive Features

The strongest SAE features associated with ROA decline were highly interpretable, but they were mostly related to occupation, industry, or standardized posting structure rather than broad recruiting-language style. Among the positively correlated features, the three strongest described front-office financial markets and risk-related work ($r = 0.527$), corporate finance and investment banking ($r = 0.435$), and retail branch banking and sales ($r = 0.379$).

The negatively correlated features showed a similar pattern. They included standardized retail posting headers ($r = -0.317$), detailed Equal Opportunity Employer language ($r = -0.265$), auto-parts retail work ($r = -0.245$), and several trucking- and cashiering-related features.

These results showed that HypotheSAEs successfully identified textual dimensions that were predictive of the firm-level label. However, many of the strongest dimensions reflected differences in the types of jobs being posted across firms, rather than whether firms under declining financial performance systematically used different recruiting language within comparable jobs.

This was an important limitation in my application. Because the label is defined at the firm level but the text is observed at the posting level, occupation and industry composition can become strong predictive signals. As a result, the most predictive SAE features do not correspond to the recruiting-language dimensions I was interested in, such as masculine versus feminine wording, emphasis on agentic versus communal traits, or stronger performance-oriented expectations.

Holdout Evaluation

I evaluated the three strongest training hypotheses on the 47 firms that were held out from all previous steps. I focused on the top three because evaluating all 20 hypotheses on 1,271 holdout postings would require more than 25,000 annotations and would be expensive and time-consuming.

Only one of the three evaluated hypotheses was statistically significant in the holdout sample. The corporate finance and investment banking feature had a separation score of 0.097 ($p = 0.006$).

and a regression coefficient of 0.096 ($p = 0.006$). The retail branch banking feature was not significant ($p = 0.635$). The strongest training feature, which represented front-office financial markets and risk-related work, did not appear in the holdout sample, so its separation score could not be estimated. Using these three evaluated hypotheses, the holdout AUROC was 0.532, which is close to random classification.

This result was consistent with what I observed in the training data. The SAE learned clear and interpretable patterns, but some of the strongest patterns were closely tied to particular occupations, industries, or firms. These patterns could be strongly correlated with the outcome in the training sample without generalizing well to firms that were not represented in the training data.

Reflection and Possible Extensions

The main thing I learned from this exercise is that HypothSAEs can find interpretable textual features, but those features are not necessarily the ones a researcher originally cares about, especially in hierarchical data. In my setting, the outcome is measured at the firm level while the text is measured at the job-posting level. This made it easy for the model to pick up differences in occupation, industry, or firm-specific posting templates. These features can be predictive of ROA decline without representing differences in recruiting language itself, such as masculine versus feminine wording or stronger performance-oriented expectations.

I also tried removing explicit job titles from the text as an additional preprocessing step, but this did not solve the problem. Many occupations are still easy to infer from the remaining posting language because responsibilities, required skills, certifications, and industry-specific terms are highly distinctive. I also drafted code for a cosine-similarity benchmark at the end (in response to 3.b. in the assignment). However, because the notebook kernel restarted late in the process, rerunning the full sequence would require more time, so I did not execute this extension and leave the code for reference.

A promising next step may be to change the comparison rather than only clean the text: identify predictive SAE features within the same occupation or industry, or residualize feature activations with respect to occupation and industry before relating them to the firm-level outcome. More generally, developing ways to separate within-group language variation from between-group compositional differences may be a direction for applying HypothSAEs to hierarchical data.

LLM Use and Time Allocation

I first used ChatGPT and Gemini to help me clarify technical concepts while reading the paper. I used Codex to help construct and check company-name aliases during text preprocessing, and to troubleshoot API limits and find a more efficient way to process the embedding requests in smaller batches. The main data preprocessing and modeling decisions, interpretation of the SAE features, evaluation of the results, and report drafting were done by myself. I also used ChatGPT to check the clarity of my explanations, particularly whether they would be understandable to CS readers given that my background is in social science. I then revised the wording myself based on its comments.

The total duration was about 7 hours. I spent about 1.5 hours reading the paper, around 3 hours on coding and implementation (including trying the review-data example, debugging the data pipeline, and running the model), and about 2.5 hours writing and revising the report.